

SUMMARY

Software Engineer at SAP Labs (BTP Core) and Master of Engineering student specializing in Cloud Computing with expertise in Cloud Architecture, DevOps, and cloud automation. Experienced in building scalable cloud solutions through hands-on projects.

EXPERIENCE

SAP Labs – BTP Core

Aug 2025 – Present

Software Engineer Intern – SAP Document Management Service (SDM) on SAP BTP enables centralized, secure handling of unstructured business content for Cloud ERP environments, ensuring seamless storage, retrieval, governance, and streamlined document lifecycles through scalable cloud architecture supporting compliance, collaboration, and efficient enterprise-wide content integration.

- Developed a multi-threaded **Python** toolkit to enable forward and reverse metadata migration between **MongoDB** and **PostgreSQL**, improving data portability and migration efficiency.
- Worked with **Cloud Foundry** on SAP BTP to deploy, manage, and monitor applications, gaining hands-on experience in cloud-native application lifecycle management.
- Self-managed **PostgreSQL** setup with **High Availability**.

PROJECTS

AIOps Agentic Self-Healing Kubernetes Platform

03/2026

Developed an AI-driven AIOps platform for automated incident detection, Root Cause Analysis (RCA), and self-healing in Kubernetes. <https://github.com/bharathac5775/aiops-agentic-platform>

- Built a **Multi-Agent System (LangGraph)** for automated monitoring, analysis, remediation, and reporting.
- Integrated **Prometheus**, **Alertmanager**, and **Loki** to enable real-time observability and log aggregation.
- Implemented **RAG-based RCA** using **ChromaDB** and LLMs via **Ollama** for intelligent troubleshooting.
- Developed policy-driven **Auto-Remediation** (restarting, scaling) using the **Kubernetes Python Client**.
- Automated build and deployment workflows using **Jenkins CI/CD** pipelines.
- Integrated persistent incident tracking and **Discord-based** alert notifications for real-time incident response.

Progressive Delivery Techniques

02/2025 – 04/2025

Implemented progressive delivery using **Kubernetes**, **Istio**, **Flagger**, and **ArgoCD** to enable automated canary deployments with zero downtime.

- Designed and executed **Canary Deployment** strategies using **Argo Rollouts** and **Flagger**.
- Automated CI pipelines using **Jenkins** and integrated **GitOps** workflows with **Argo CD** to deploy Helm chart updates.
- Improved observability using **Kiali**, **Prometheus**, **Grafana**, **EFK** and **Jaeger**.

Designing Secure and High-Performance 3-tier Web Application on AWS

08/2024 – 12/2024

Designed and deployed a scalable **3-tier web application on AWS** with high availability and fault tolerance.

- Hosted a static website on **Amazon S3** and built a dynamic application on **Amazon EC2**.
- Designed a secure **VPC** with public and private subnets to isolate web and database layers.
- Migrated the database to managed **Amazon RDS** and implemented serverless reporting using **AWS Lambda**.
- Implemented scalability and reliability using **Auto Scaling Groups**, **Elastic Load Balancer**, **Route 53**, monitored with **CloudWatch**, and automated infrastructure using **CloudFormation**.

SKILLS

Programming Languages and Database

C programming | Data Structures | Java (Basics) | Python (Basics) | SQL Databases

DevOps Tools

Git | GitHub | Docker | Kubernetes | Jenkins | CircleCI | GitLab | ArgoCD | Flagger | Istio service Mesh

Ansible | Terraform | Prometheus | EFK stack | Loki | Jaeger | SAP-Cloud Foundry | SAP-Kyma | AIOps

AWS Cloud Management Service

Compute: EC2 | ECS | EKS | Lambda | Beanstalk Storage: S3 | EBS | EFS Database: RDS | DynamoDB
Networking and CDN: VPC | Route53 | Load Balancer | CloudFront Cloud Security | Systems Manager
IaC: CloudFormation Monitoring: CloudWatch | CloudTrail

Operating System and Networking

Linux | Windows Computer Networks

EDUCATION

Master of Engineering Manipal School of Information Sciences (MSIS) Manipal Academy of Higher Education (MAHE), Manipal	GPA: 9.56 / 10.0 07/2024 – Present
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Bachelor of Engineering NMAM Institute of Technology (NITTE) Visvesvaraya Technological University, Belagavi	GPA: 8.8 / 10.0 12/2020 – 03/2024
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CERTIFICATION

AWS Academy Cloud Foundations	AWS Academy Cloud Architecting
AWS Academy Cloud Security and Builder	AWS Academy Cloud Operations
DevOps and AI on AWS (Coursera)	KodeKloud Certificates
Web Development Bootcamp (Udemy)	

LANGUAGES

Kannada English Hindi